

Frequency, $f = \frac{1}{T}$ or $T = \frac{1}{f}$

Q. Given the frequencies below, calculate the corresponding period.

① 24 Hz

$$T = \frac{1}{24} \text{ s} = 0.04166 = \underline{\underline{41.6 \text{ ms}}}$$

② 8 MHz

$$T = \frac{1}{8 \text{ M}} = \frac{1}{8 \times 10^6} = 0.125 \times 10^{-6} = \underline{\underline{125 \text{ ns}}}$$

Q. Given the periods calculate the frequencies.

① 5 s

$$f = \frac{1}{T} = \frac{1}{5} = \underline{\underline{0.2 \text{ Hz}}}$$

② ~~12 ns~~ ~~12 μs~~ μs

$$f = \frac{1}{12 \times 10^{-6}} = \underline{\underline{83.33 \times 10^3 \text{ Hz}}} = \underline{\underline{83.33 \text{ kHz}}}$$

* Phase

→ Describes position of waveform relative to time 0.

* Composite signal.

— made of many simple sine waves.

* Digital Signal

→ Value 1 — positive voltage
Value 0 — Zero voltage.

→ $\log_2 4 = 2$ bits

→ Bit rate — no. of bits per second.

→ $100 \times 24 \times 80 \times 8 = 1.536 \text{ Mbps}$.

→ Bit Length = propagation speed $\times$ bit duration
— distance one bit occupy on the transmission medium.

Q. What is the bit rate for each of the following signals?

$$T = \frac{\text{data bits}}{\text{data rate}}$$

$$\text{Bit rate} = \frac{1}{\text{Bit duration}}$$

a) A signal in which 1 bit last 0.001 second, calculate bit rate.

$$\rightarrow \text{Bit rate} = \frac{1}{0.001} = \underline{1000 \text{ bps}}$$

b) A signal in which 1 bit last 2 ms.

$$\text{Bit rate} = \frac{1}{2 \times 10^{-3}} = 0.5 \times 10^3 = \underline{500 \text{ bps}}$$

c) A device is sending out data at the rate of 1000 bps.

a) How long does it take to send 10 bits.
→ $\frac{10 \times 10}{1000} = \frac{100}{1000} = 0.1 \text{ s} = \underline{10 \text{ ms}}$

b) How long does it take to send out a single character?

$$\rightarrow \frac{8}{1000} = 8 \times 10^{-3} = \underline{8 \text{ ms}}$$

c) 100000 characters

$$\frac{100,000 \times 8}{1000} = \underline{800 \text{ s}}$$

- What is sent is not what is received.
- 3 causes of impairment are attenuation, distortion and noise.

* Attenuation and Amplification

- Means loss of energy.
- To compensate - we need amplification

• Decibel

- Measures relative strength of 2 signals or one signal at 2 different point

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$

P_1, P_2 - power of signal at pt 1 and 2.

⇒ negative - Signal is attenuated

⇒ positive - " Amplified

- Q. A signal travels ^{through} ~~to~~ an amplifier and its power increase 10 times. calculate signal strength

$$\rightarrow 10 \log_{10} \frac{10P_1}{P_1} = \underline{\underline{10 \text{ dB.}}}$$

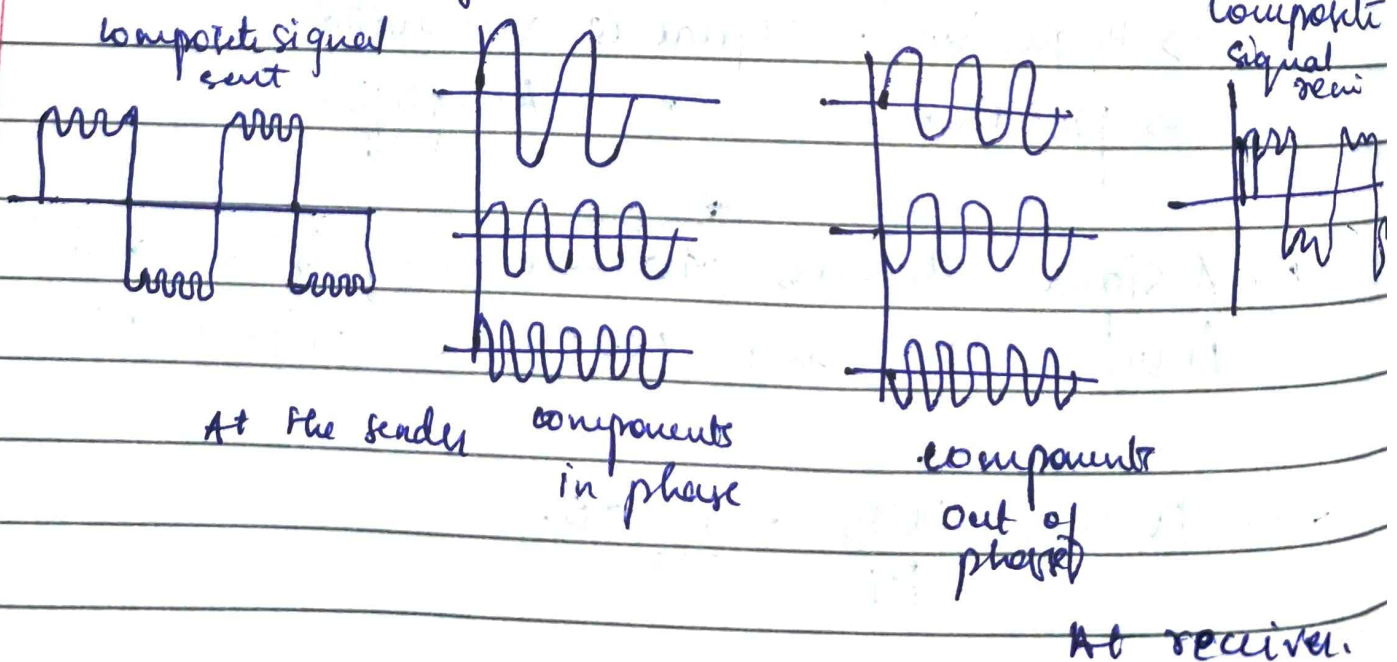
Q. A signal travel from A $\rightarrow$ B. At A, power is 100W, at B power is 90W. What is attenuation in dB?

$$10 \log_{10} \frac{90}{100} = 10 \log_{10} 0.9 = 10(-0.046) \\ = \underline{\underline{-0.46 \text{ dB}}}$$

$$Q. 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{110}{100} = 10 \log_{10} 1.1 \\ = \underline{\underline{0.41 \text{ dB}}}$$

* Distortion

- Signal changes its form or shape.
- Each signal component has its own propagation speed through a medium.



Q. Attenuation of signal is -10 dB. What is signal power if it was originally 5 W. final

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$

$$-10 = 10 \log_{10} \frac{P_2}{5}$$

$$-1 = \log_{10} P_2 - \log_{10} 5$$

$$\log_{10} P_2 = -1 + \log_{10} 5 = -0.3$$

$$\log_{10} P_2 = -0.3$$

$$P_2 = 0.5 \text{ W}$$

✓

or $-1 = \log_{10} \frac{P_2}{5}$

$$10^{-1} = \frac{P_2}{5} \quad (\text{Rewrite the base 10 log in its exponential form } \rightarrow 10^y = x)$$

$$P_2 = 0.5 \text{ W}$$

$$\Rightarrow 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5}{5} = 10 \log_{10} 10^{-1} = -10 \text{ dB}$$

★ $SNR_{dB} = 10 \log_{10} SNR$

Q. ~~SNR~~ Power of signal is 10 mW and power of noise is $1 \mu\text{W}$. What are the values of SNR and SNRdB?

$$\rightarrow SNR = \frac{10 \text{ mW}}{1 \mu\text{W}} = \frac{10 \times 10^{-3}}{1 \times 10^{-6}} = 10 \times 10^3 = 10^4$$

$$SNR_{dB} = 10 \log_{10} SNR$$

$$= 10 \log_{10} 10^4$$

$$= 40 \text{ dB}$$

* Values of SNR and SNRdB for noiseless channel are

$$SNR = \infty, SNR_{dB} = \infty$$

* Data Rate Limits

→ Data rate depends on 3 factors

- bandwidth available
- level of signals we use
-

★ Nyquist bit rate formula.

$$\text{★ Bitrate} = 2 \times \text{bandwidth} \times \log_2 L$$

- L: no of signal levels used to represent data.
- bps.

Q. Consider a noiseless channel with bandwidth of 3000 Hz + transmitting a signal with 2 signal levels. the max bit rate is —

$$\rightarrow \text{Bitrate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps.}$$

Q. We need to send 265 kbps over a noiseless channel with bandwidth of 20 kHz. How many signal levels do we need?

$$\text{Bitrate} = 2 \times \text{Bandwidth} \times \log_2 L$$

$$265 \text{ kbps} = 2 \times 20 \text{ kHz} \times \log_2 L$$

$$\log_2 L = \frac{265 \times 10^3}{2 \times 20 \times 10^3} = \frac{265}{40} = \frac{53}{8}$$

$$\log_2 L = \frac{53}{8}$$

$$L = 2^{\frac{53}{8}} = 98.7 \text{ levels.}$$

★ Noisy channel: Shannon Capacity

$$\text{★ capacity} = \text{bandwidth} \times \log_2 (1 + \text{SNR}) \text{ (bps)}$$

Q. Consider extremely noisy channel in which the value of the signal-to-noise ratio is almost 0. For this channel the capacity C is —

$$C = B \log_2 (1 + 0) = 0 \text{ bps.}$$

Q. A telephone line normally has bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. Calculate channel capacity.

$$C = \text{bandwidth} \times \log_2 (1 + \text{SNR})$$

$$= 3000 \times \log_2 (1 + 3162)$$

$$= 3000 \times \log_2 (3163)$$

$$= 34881.23$$

$$= 3.488123 \times 10^4 \text{ bps.}$$

$$= 3000 \times \frac{\log_{10} 3163}{\log_{10} 2} \text{ (change of base rule)}$$

$$= 34881.23 \text{ bps}$$

Q. Assume that $\text{SNR}_{\text{dB}} = 36$ and channel bandwidth is 2 MHz. Calculate the theoretical channel capacity.

$$\text{SNR}_{\text{dB}} = 36, B = 2 \text{ MHz}$$

$$C = B \log_2 (1 + \text{SNR})$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

$$\frac{36}{10} = \log_{10} \text{SNR}$$

$$10^{3.6} = \text{SNR}$$

$$\text{SNR} = 3981$$

$$C = 2 \times 10^6 \times \log_2 (1 + 3981)$$

$$= 2 \times 10^6 \times \log_2 (3982)$$

$$= 2 \times 10^6 \times \frac{\log_{10} 3982}{\log_{10} 2}$$

$$= 23918555.01$$

$$= \underline{\underline{23.918555 \text{ Mbps}}} \approx \underline{\underline{23.92 \text{ Mbps}}}$$

Q. We have a channel with a 1 MHz bandwidth. The SNR of this channel is 63. What are the appropriate bit rate and signal level?

$$C = B \log_2 (1 + \text{SNR}) \quad (\text{to find upper limit}).$$

$$= 10^6 \log_2 (1 + 63)$$

$$= 10^6 \log_2 64$$

$$C = \underline{\underline{6 \text{ Mbps}}}$$

For better performance we choose something lower, 4 Mbps for example.

Then use Nyquist formula,

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$$

$$\Rightarrow \underline{\underline{L = 4}}$$

Performance

* Bandwidth

Throughput

Latency (Delay)

Bandwidth-Delay Product

Jitter

* Performance :-

① Bandwidth (B)

→ Range of frequencies in a composite signal or range of frequencies.

② Throughput (T)

→ Measure of how fast we can actually send data through a network.

→ T always less than B.

Q. A network with bandwidth of 10 Mbps can pass only an avg of 12,000 frames per minute with each frame carrying an avg of 10,000 bits. What is the throughput of this network?

$$\rightarrow \text{Throughput} = (12,000 \times 10,000) / 60 = \underline{2 \text{ Mbps.}}$$

③ Latency

→ How long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

* Latency = propagation time + transmission + queuing + processing delay

* 1 byte = 8 bits.

• Propagation time : measures time required for a bit to travel from source to destination.

* $\Rightarrow \text{Distance} / (\text{Propagation Speed})$.

Q. What is the propagation time if distance b/w 2 points is 12,000 km? Assume propagation speed to be $2.4 \times 10^8 \text{ m/s}$ in cable?

$$\rightarrow \text{Propagation time} = (12,000 \times 1000) / (2.4 \times 10^8) = \underline{50 \text{ ms.}}$$

• Transmission time:

* $\Rightarrow (\text{Message Size}) / \text{bandwidth}$.

Q. What are propagation time and transmission time for 2.5 KB message if bandwidth of network is 1 Mbps? Assume the distance b/w sender and receiver is 12,000 km and light travels at $2.4 \times 10^8 \text{ m/s}$.

$$\rightarrow \text{Propagation time} = \frac{12,000 \times 1000}{2.4 \times 10^8} = \underline{50 \text{ ms}}$$

$$\text{Transmission time} = \frac{2.5 \text{ KB} \times 8}{1 \text{ Mbps}} = \frac{20 \times 10^3}{10^6} = \underline{20 \mu\text{s}}$$

(bits from not byte. $\Rightarrow 0.02 \text{ ms}$)

Q. What are propagation time and transmission time for 5MB message if bandwidth of network is 1 Mbps? Assume distance b/w sender and receiver is 120000 and light speed is 2.4×10^8 m/s.

$$\rightarrow PT = \frac{12,000 \times 1000}{2.4 \times 10^8} = \underline{\underline{50 \text{ ns}}}$$

$$TT = \frac{5 \times 10^6 \times 8}{1 \times 10^6} = \underline{\underline{40 \text{ s}}}$$

• Queueing Queueing time:-

④ Bandwidth-Delay product

* Digital Transmission:-

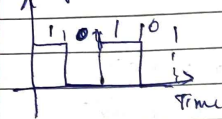
→ Info needs to be converted to either a digital or analog signal before transmission.

→ A computer network is designed to send info from one point to another.

• Digital to digital conversion.

→ line coding, block coding, scrambling

Voltage



$$\frac{1}{2} V^2 \Rightarrow \text{Power}$$

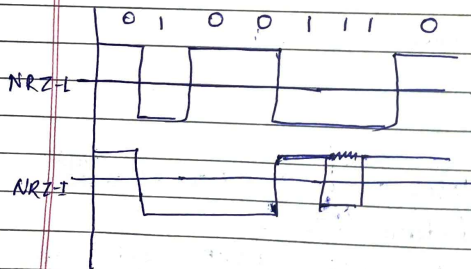
③ ~~Double~~ Double power required.

★ Polar scheme

- Voltage on both side of ~~vertical~~ time axis.
- Voltage level 0 — positive
1 — negative

NRZ

↳ 2 version — NRZ-L, NRZ-I



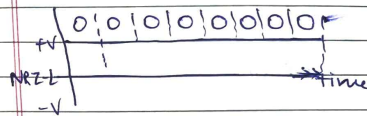
28

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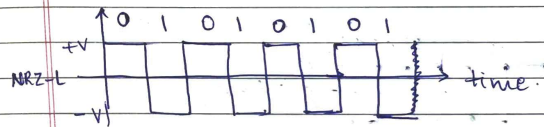
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Q. Draw the graph for following data streams using NRZ-L and scheme.

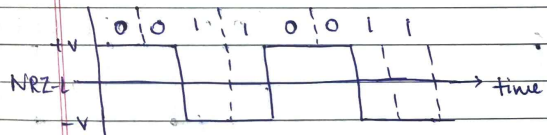
1) 00000000



2) 01010101

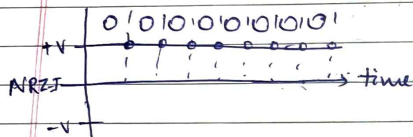


3) 00110011

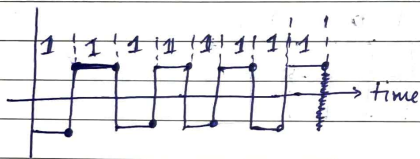


Q Draw the graph for the waveform for following data streams. using NRZ-I

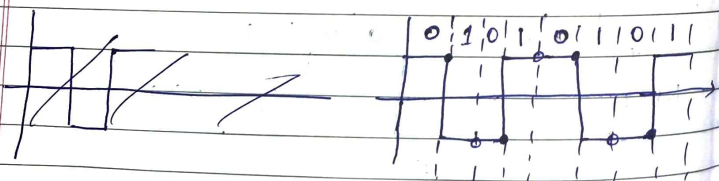
1) 00000000



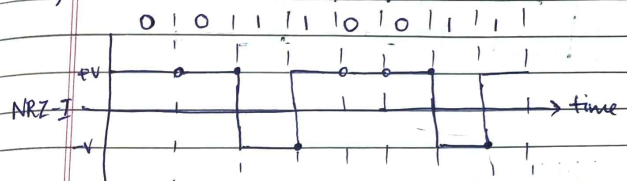
2) 11111111



3) 01010101

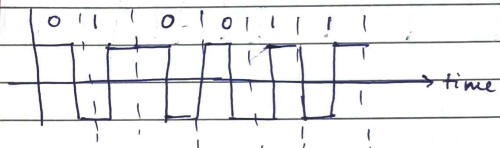


Q. 00110011

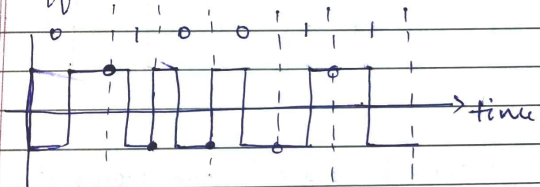


Manchester

○ → Next bit 1
● → Next bit 0



Differential manchester.

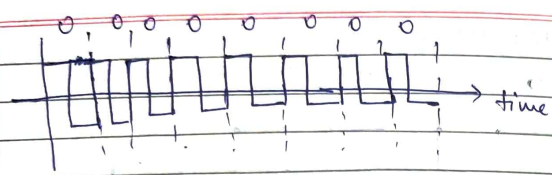


Q. Draw waveform for Manchester.

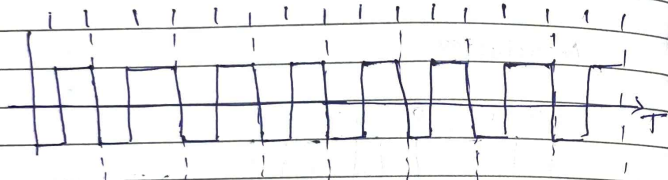
1) 00000000

analog: 0's - 7 1's.

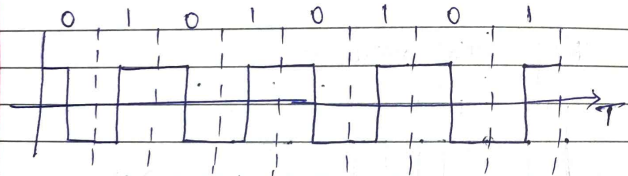
classmate
Date _____
Page _____



2) 11111111



3) 01010101



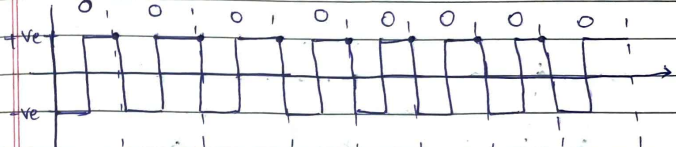
* Biphase: Manchester and differential manchester :-

Q. Draw the waveform for differential manchester encoding.

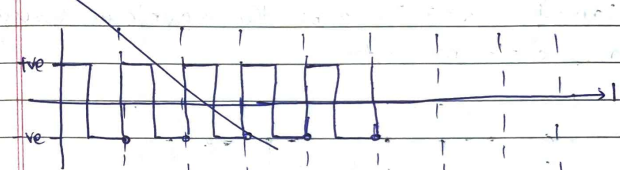
(Inversion: Next bit is 0). 0

classmate
Date _____
Page _____

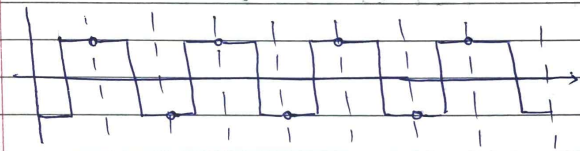
1) 00000000



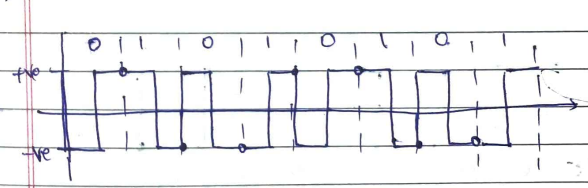
2) 11111111



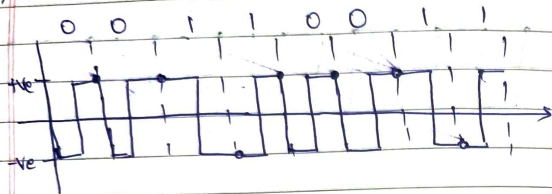
2)



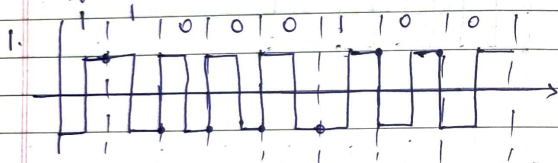
3) 01010101



4) 00110011

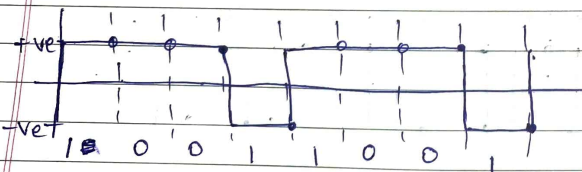


Q. Identify data stream on the waveform using differential manchester



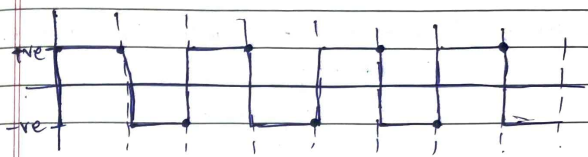
→ 11000100

Q. Write 8 bit data stream for below graph using NRZ-I.



⇒ 10011001

Q. NRZ-I - 11111111



★ Transmission Media:-

- located below the physical layer and are directly controlled by physical layer.
- defined as anything that can carry info from source to destination.
- Divided into guided and unguided.

• Unguided Media: Wireless

- transport electromagnetic waves without using physical conductor.

• Guided Media

- provide a conduit from one device to another.
- twisted pair, coaxial, fibre optic.

★ Block coding

→ message divided into blocks each of k -bits → data words.

→ we add r redundant bits → $n = k + r$.
↓
codewords.

★ Minimum Hamming Distance for error detection

→ errors.
★ $d_{\min} = S + 1$

Q. $d(000, 011)$ is 2 bec $000 \text{ XOR } 011 = 011$ (2 1's)

Q. $d(10101, 11110) = 3$

$$\begin{array}{r} 10101 \\ \text{XOR } 11110 \\ \hline 01011 \end{array} \quad (3 \text{ 1's})$$

★ Linear Block Codes:-

Data words	Codewords	Data words	Codewords
00	000	10	101
01	011	11	110

2 valid codeword XOR → codeword.

★ Parity-check codes:-

- This code is a linear block code.
- k bit ~~keyword~~ dataword changed to n bit codeword.

Q. dataword

1011

↓

10111

Codeword

$$\Rightarrow (1+0+1+1+1) \times 2 = 0 \rightarrow \text{Syndrome.}$$

★ Cyclic code:-

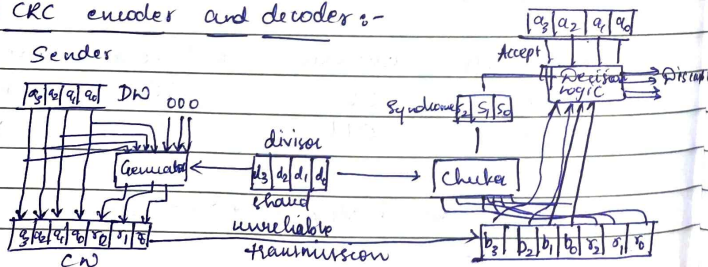
- If codeword is cyclically shifted the result is cw.

→ Ex:

1011000 (left shift)
0110001

★ CRC encoder and decoder:-

Sender



Q. Dataword: 1001

$$\begin{array}{r} 1010 \rightarrow \text{Discard} \\ 1001 \mid 1001000 \\ \underline{1001} \\ 0100 \\ \underline{0000} \\ 1000 \\ \underline{1011} \\ 0110 \\ \underline{0000} \\ 110 \rightarrow \text{remainder} \end{array}$$

CW = 1001110

DW redundant bits.

Receiver side

$$\begin{array}{r} 1010 \\ 1001 \mid 1001110 \\ \underline{1011} \\ 0101 \\ \underline{0000} \\ 1011 \\ \underline{1011} \\ 0000 \\ \underline{0000} \\ 000 \rightarrow \text{remainder is 0 so valid.} \end{array}$$

Error case

$$\begin{array}{r}
 1000 \\
 1011 \overline{) 1011110} \\
 \underline{1011} \\
 0001 \\
 \underline{0000} \\
 0011 \\
 \underline{0000} \\
 0110 \\
 \underline{0000} \\
 110
 \end{array}$$

remainder is not 0 so error

Q. Create codeword for DW 0011 through CRC.
Consider divisor 1011.

$$\begin{array}{r}
 0011 \\
 1011 \overline{) 0011000} \\
 \underline{0000} \\
 0110 \\
 \underline{0000} \\
 1100 \\
 \underline{1011} \\
 1110 \\
 \underline{1011} \\
 101
 \end{array}$$

CW: 0011101

Q. Given DW 10100111, divisor: 1011 through CRC codeword at the sender side using binary division (discard)

$$\begin{array}{r}
 10011011 \\
 1011 \overline{) 101001110000} \\
 \underline{1011} \\
 000111 \\
 \underline{0000} \\
 001111 \\
 \underline{0000} \\
 111111 \\
 \underline{1011} \\
 10001 \\
 \underline{1011} \\
 01100 \\
 \underline{0000} \\
 00000
 \end{array}$$

CW: 1010011110101

$$\begin{array}{r}
 11000 \\
 1011 \overline{) 11110} \\
 \underline{1011} \\
 10010 \\
 \underline{1011} \\
 10101
 \end{array}$$

∴ CW = 1010011110101

Q Assuming ~~given~~ even parity find parity bit for each of the following data units.

a) 1001011 $\rightarrow$ 10010110
(1+0+0+1+0+1+1)%2 = 0 = r_0 .

b) 0001100
 $r_0 = 0$

c) 1000000
 $r_0 = 1$

Q. CRC If DW is 5 bits and CW is 8 bits. How many 0 are needed to be added to DW to make divisible? What is size of remainder, divisor?

$\rightarrow$ 3. Size of remainder = 3 (No of 0).
Size of divisor = 4

In block code a DW is 20 bits and CW is 25 bits. What are the values of k, r, n .

$\rightarrow k=20, n=25, r=5$.

Q If we want to detect ² bit errors. What should be the min hamming distance

$\rightarrow d = 2 + 1 = 3$ bits.

Q What is the hamming distance for each of the following CW.

$\rightarrow d(10000, 00000) = 3 + 1 = 1$

$d(10101, 10000) = 2 + 1 = 2$

$d(00000, 11111) = 5 + 1 = 5$

Perform XOR of both, count 1's.

Q. Using the code in table shown what is DW if each of the following CW. Is detected.

a) 0101 - No DW $\%2 \neq 0$.

b) 11111 - No DW $\%2 \neq 0$

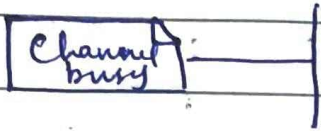
c) 00000 - 00000 $\%2 = 0$.

d) 11011 - 11011 $\%2 = 0$

Return to 0

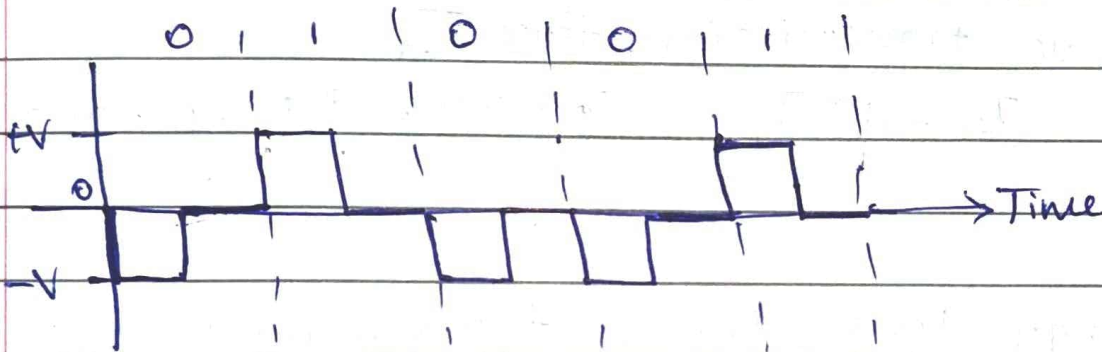
→ 0 means -ve start 
→ 1 means +ve start 

• 1 - Persistent method.

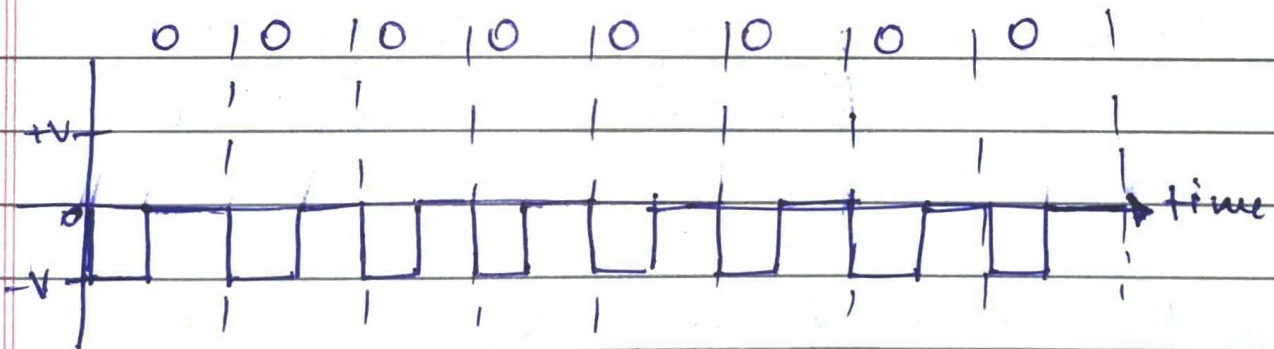


Q. Draw the waveform for the given data stream using return to 0 time coding technique.

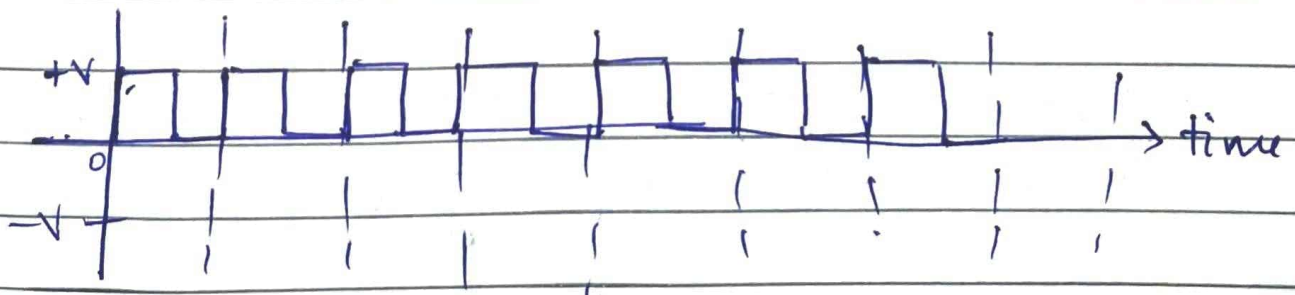
a) ~~1111~~ 01001



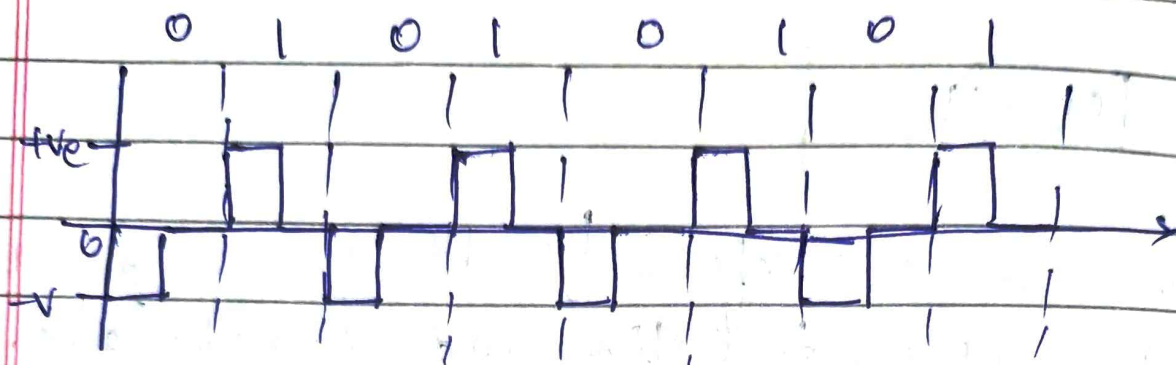
b) 010000000



c) 111111111



d) 01010101

★ Frame transmission time T_{fr}

$$T_{fr} = 2 * T_p \quad T_p - \text{max propagation time.}$$

★ Energy level during transmission, idleness or collision :-

→ 3 values: zero, normal, abnormal.

↓	↓	↓
idle	success	collision

★ Link layer Addressing:-

* Unicast Address - one to one communication

* Broadcast address - one to all communication
— 12 hexadecimal digits.